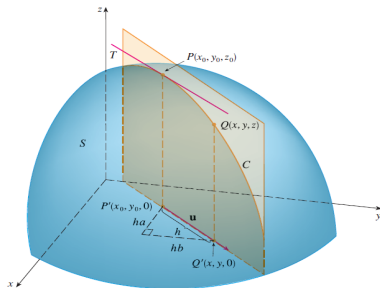
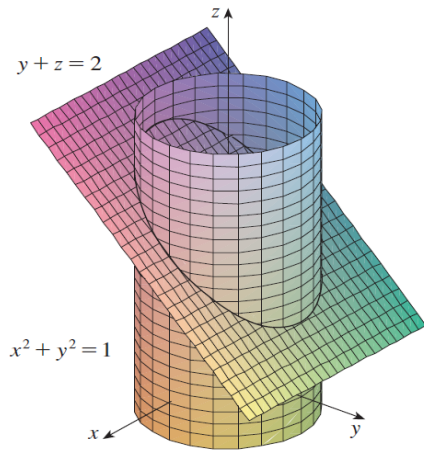


高等微积分 (2)

第 17 章: 曲线曲面积分



提要: 第 2 型曲线积分等



复习: 第一类曲线积分

1. 第一类曲线积分 $\int_L f(x, y, z) d\ell$ 的定义:
形式上与 $[a, b]$ 上的定积分完全相同, 但却要将闭区间 $[a, b]$ 换成空间曲线 L .
2. 若 L 为分段光滑曲线, 而函数 f 为分段连续, 则第一类曲线积分 $\int_L f(x, y, z) d\ell$ 存在.
3. 如果 $L \subset \mathbb{R}^2$, 则曲线积分 $\int_L f(x, y) d\ell$ 为柱面 $z = f(x, y), (x, y) \in L$ 的面积.

- 4. 第一类曲线积分具有与 $[a, b]$ 上的定积分完全类似的性质.
- 5. 第一类曲线积分与曲线的方向无关.
- 6. 空间曲线 L 的质心 $(\bar{x}, \bar{y}, \bar{z})$:

$$\begin{aligned}\bar{x} &= \frac{1}{M} \int_L x \rho(x, y, z) d\ell, & \bar{y} &= \frac{1}{M} \int_L y \rho(x, y, z) d\ell, \\ \bar{z} &= \frac{1}{M} \int_L z \rho(x, y, z) d\ell, & M &= \int_L \rho(x, y, z) d\ell,\end{aligned}$$

其中 ρ 为质量分布的密度函数.

复习: 第一类曲线积分的计算

设分段光滑曲线 L 的参数方程为

$$\begin{cases} x = x(t), \\ y = y(t), \\ z = z(t), \end{cases} \quad t \in [\alpha, \beta],$$

则第一类曲线积分 $\int_L f(x, y, z) d\ell$ 等于

$$\int_{\alpha}^{\beta} f(x(t), y(t), z(t)) \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt.$$

特别地, 若曲线 $L \subset \mathbb{R}^2$ 由方程

$$y = y(x), \quad x \in [a, b]$$

给出, 则我们有

$$\int_L f(x, y) d\ell = \int_a^b f(x, y(x)) \sqrt{1 + (y'(x))^2} dx.$$

同样地, 若曲线 $L \subset \mathbb{R}^2$ 由方程

$$x = x(y), \quad y \in [c, d]$$

给出, 则我们有

若 L 在极坐标系下的方程为

$$\rho = \rho(\varphi), \quad \varphi \in [\alpha, \beta],$$

则 L 在直角坐标系下的参数方程为

$$x = \rho(\varphi) \cos \varphi, \quad y = \rho(\varphi) \sin \varphi,$$

于是曲线 L 的弧微分为

$$\begin{aligned} d\ell &= \sqrt{(x'(\varphi))^2 + (y'(\varphi))^2} d\varphi \\ &= \sqrt{(\rho'(\varphi) \cos \varphi - \rho(\varphi) \sin \varphi)^2 + (\rho'(\varphi) \sin \varphi + \rho(\varphi) \cos \varphi)^2} d\varphi \\ &= \sqrt{(\rho'(\varphi))^2 + \rho(\varphi)^2} d\varphi \end{aligned}$$

若曲线 L 由隐函数方程组

$$\begin{cases} \varphi_1(x, y, z) = 0, \\ \varphi_2(x, y, z) = 0, \end{cases}$$

给出, 利用隐函数定理来局部求解上述方程组, 由此得到曲线 L 的分段的参数表示, 随后再对每段分别利用前面的公式计算.

复习: 第一类曲面积分

- a. 第一类曲面积分 $\iint_S f(x, y, z) d\sigma$ 的定义:
形式上与二重积分相同, 但要将二维的坐标平行体换成曲面 S , 故二者性质完全类似.
- b. 若 S 为分片光滑曲面, 而 f 为分片连续函数, 则 $\iint_S f(x, y, z) d\sigma$ 存在.
- c. 如果 $S \subset \mathbb{R}^2$, 则 $\iint_S f(x, y) d\sigma$ 为 S 上且介于曲面 $z = 0$ 与 $z = f(x, y)$ 之间的立体体积.

复习: 第一类曲面积分的计算

设分片光滑曲面 S 的参数方程为

$$\begin{cases} x = x(u, v), \\ y = y(u, v), \quad (u, v) \in D, \\ z = z(u, v), \end{cases}$$

其中 $D \subset \mathbb{R}^2$ 为 Jordan 可测集, 则面积微元为

$$d\sigma = \sqrt{EG - F^2} du dv,$$

其中

$$E = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2,$$

$$G = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2,$$

$$F = \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \frac{\partial z}{\partial v}.$$

于是我们有

$$\iint_S f(x, y, z) d\sigma = \iint_D f(x(u, v), y(u, v), z(u, v)) \sqrt{EG - F^2} du dv$$

特别地, 若曲面 S 由方程

$$z = z(x, y), \quad (x, y) \in D$$

给出, 则我们有

$$\iint_S f(x, y, z) d\sigma = \iint_D f(x, y, z(x, y)) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy$$

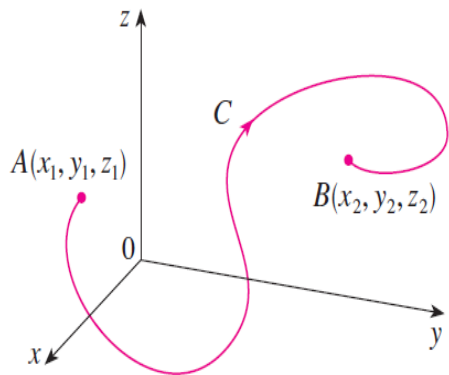
当曲面 S 由方程 $x = x(y, z)$ 或方程 $y = y(x, z)$

给出时, 我们也有类似的公式.

17.2. 第二类曲线积分

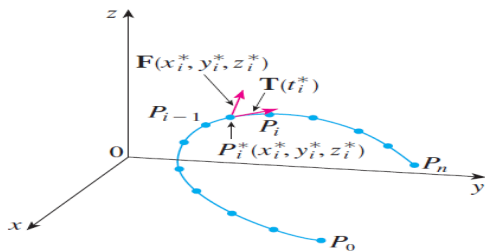
定义 1. 设 $L \subset \mathbb{R}^3$ 为空间曲线, 它的起点为 A , 它的终点为 B , 而 $\vec{F} = (F_1, F_2, F_3) : L \rightarrow \mathbb{R}^3$ 为向量值函数. 即

$$\vec{F}(x, y, z) = \left(F_1(x, y, z), F_2(x, y, z), F_3(x, y, z) \right),$$
$$(x, y, z) \in L \subset \mathbb{R}^3.$$



对任意整数 $n \geq 1$, 我们将曲线 L 分割成 n 小段: $\widehat{P_0 P_1}$, $\widehat{P_1 P_2}$, $\dots$, $\widehat{P_{n-1} P_n}$, 其中 $P_0 = A$, $P_n = B$. 我们记

$$P_i = (x_i, y_i, z_i) \quad (0 \leq i \leq n), \quad \vec{\ell} = (x, y, z),$$



并在每个小段 $\widehat{P_{i-1}P_i}$ 上取点 $X_i = P_i^* = (x_i^*, y_i^*, z_i^*)$. 令

$$d = \max_{1 \leq i \leq n} |\overline{P_{i-1}P_i}|,$$

并称之为分割的步长. 记 $P_{i-1} = (x_{i-1}, y_{i-1}, z_{i-1})$, 则

$$\overrightarrow{P_{i-1}P_i} = (x_i - x_{i-1}, y_i - y_{i-1}, z_i - z_{i-1}).$$

定义 (若极限存在)

$$\begin{aligned} & \int_{L(\widehat{AB})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} \\ &= \lim_{d \rightarrow 0} \sum_{i=1}^n \vec{F}(X_i) \cdot \overrightarrow{P_{i-1}P_i} \\ &= \lim_{d \rightarrow 0} \sum_{i=1}^n \left(F_1(X_i)(x_i - x_{i-1}) + F_2(X_i)(y_i - y_{i-1}) + F_3(X_i)(z_i - z_{i-1}) \right) \end{aligned}$$

并且称之为向量值函数 $\vec{F}$ 沿曲线 L 由点 A 到点 B 的第二类曲线积分.

注

- (a) 上述极限存在意味着: $\exists a \in \mathbb{R}$ 使得 $\forall \varepsilon > 0$, $\exists \delta > 0$ 使得当 $d < \delta$ 时, 我们均有

$$\left| \sum_{j=1}^n \vec{F}(X_j) \cdot \overrightarrow{P_{j-1}P_j} - a \right| < \varepsilon.$$

此时我们将 a 记作 $\int_{L(\widehat{AB})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell}$.

- (b) 若 L 为分段光滑曲线 (也即 L 可分成有限多段, 每段均有连续可导的参数表示), 而 $\vec{F}$

(c) 由定义立刻可知

$$\begin{aligned} \int_{L(\widehat{AB})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} &= \int_{L(\widehat{AB})} F_1(x, y, z) dx \\ &+ \int_{L(\widehat{AB})} F_2(x, y, z) dy + \int_{L(\widehat{AB})} F_3(x, y, z) dz. \end{aligned}$$

出于简化, 我们也将右边记作

$$\int_{L(\widehat{AB})} F_1(x, y, z) dx + F_2(x, y, z) dy + F_3(x, y, z) dz.$$

(c) 若 $L \subset \mathbb{R}^2$ 为平面曲线, 则

第二类曲线积分的性质

第二类曲线积分的性质可分为两类:

一类只涉及到被积函数, 这样的性质与定积分相应的性质的类似; 另外一类涉及积分路径.

(1) 路径的有向性:

$$\int_{L(\widehat{AB})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} = - \int_{L(\widehat{BA})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell}.$$

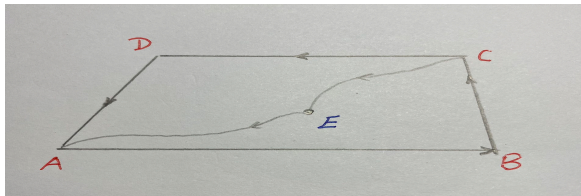
(2) 对路径的可加性: 设 $A_1, A_2, A_3 \in L$, 则

$$\begin{aligned} & \int_{L(\widehat{A_1 A_3})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} \\ &= \int_{L(\widehat{A_1 A_2})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} + \int_{L(\widehat{A_2 A_3})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell}. \end{aligned}$$

若 L 为封闭曲线, 规定其逆时针方向为正向, 此时将第二类曲线积分记作 $\oint_{L^+} \vec{F}(\vec{\ell}) \cdot d\vec{\ell}$. 考虑曲线

$L^+ : ABCDA$, $L_1^+ : ABC EA$, $L_2^+ : AECDA$, 则我们有

$$\oint_{L^+} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} = \oint_{L_1^+} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} + \oint_{L_2^+} \vec{F}(\vec{\ell}) \cdot d\vec{\ell}.$$



第二类曲线积分的计算

设分段光滑曲线 L 的参数方程为

$$\begin{cases} x = x(t), \\ y = y(t), \\ z = z(t), \end{cases} \quad t \in [\alpha, \beta],$$

其中起点 A 和终点 B 所对应的参数分别为 α, β , 则

$$\begin{aligned}
& \int_{\widehat{L(AB)}} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} \\
&= \int_{\widehat{L(AB)}} F_1(x, y, z) dx + \int_{\widehat{L(AB)}} F_2(x, y, z) dy + \int_{\widehat{L(AB)}} F_3(x, y, z) dz \\
&= \int_{\alpha}^{\beta} F_1(x(t), y(t), z(t)) x'(t) dt + \int_{\alpha}^{\beta} F_2(x(t), y(t), z(t)) y'(t) dt \\
&\quad + \int_{\alpha}^{\beta} F_3(x(t), y(t), z(t)) z'(t) dt.
\end{aligned}$$

特别地, 如果 $L \subset \mathbb{R}^2$ 为平面曲线并且其方程为 $y = y(x)$, 而 A, B 的横坐标分别为 a, b , 则

$$\int_{\widehat{L(AB)}} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} = \int_a^b F_1(x, y(x)) dx + \int_a^b F_2(x, y(x)) y'(x) dx.$$

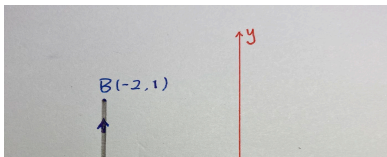
例 1. 计算下列积分

$$I_1 = \int_{L(\widehat{AB})} ((x+2)^2 - y^2) dx - 2(x+2)y dy,$$

$$I_2 = \int_{L(\widehat{AB})} 2(x+2)y dx + ((x+2)^2 - y^2) dy,$$

其中积分路径 L 是从 $A(-2, 0)$ 到 $B(-2, 1)$ 的有向直线段.

解:



路径 L 的方程为 $x = -2$ ($0 \leq y \leq 1$). 故

$$I_1 = \int_0^1 0 \mathrm{d}y = 0, \quad I_2 = \int_0^1 (-y^2) \mathrm{d}y = -\frac{1}{3}.$$

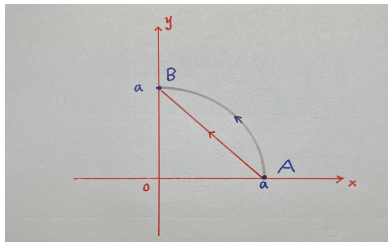
例 2. 计算 $I = \int_{L(\widehat{AB})} ydx + xdy$, 其中

(1) 路径 L 是圆弧 $x^2 + y^2 = a^2$ ($y \geq 0$, $a > 0$),

由点 $A(a, 0)$ 到点 $B(0, a)$.

(2) 路径 L 为直线段 $\overrightarrow{AB}$.

解:



(1) 路径 L 的参数方程为

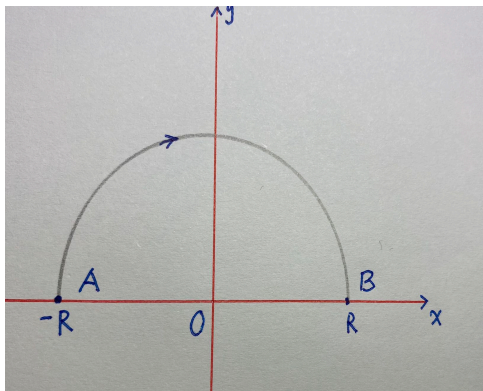
$$\begin{cases} x = a \cos \varphi, \\ y = a \sin \varphi, \end{cases} \quad \left(0 \leq \varphi \leq \frac{\pi}{2}\right).$$

于是我们有

$$\begin{aligned} I &= \int_{L(\widehat{AB})} ydx + xdy \\ &= \int_0^{\frac{\pi}{2}} a \sin \varphi d(a \cos \varphi) + (a \cos \varphi) d(a \sin \varphi) \\ &= \int_0^{\frac{\pi}{2}} a^2 (-\sin^2 \varphi + \cos^2 \varphi) d\varphi \\ &= \int_0^{\frac{\pi}{2}} a^2 \cos 2\varphi d\varphi = \frac{1}{2} a^2 \sin 2\varphi \Big|_0^{\frac{\pi}{2}} = 0. \end{aligned}$$

例 3. 计算 $\int_{L(\widehat{AB})} xdy - ydx$, 其中 L 为上半圆
 $x^2 + y^2 = R^2$ ($y \geq 0$), $A = (-R, 0)$, $B = (R, 0)$.

解:



方法 1. 曲线 L 的参数方程为

$$\begin{cases} x = R \cos t, \\ y = R \sin t, \end{cases} \quad t \in [0, \pi].$$

由于 A, B 分别对应于参数 $t = \pi, 0$, 于是

$$\begin{aligned} \int_{\widehat{L(AB)}} x dy - y dx &= \int_{\pi}^0 R \cos t d(R \sin t) - R \sin t d(R \cos t) \\ &= \int_{\pi}^0 R^2 dt = -\pi R^2. \end{aligned}$$

方法 2. 曲线 L 方程为

$$y = \sqrt{R^2 - x^2}, \quad x \in [-R, R],$$

并且点 A 对应于 $x = -R$, 点 B 对应于 $x = R$,
由此我们立刻可得

$$\begin{aligned} \int_{\widehat{L(AB)}} x dy - y dx &= \int_{-R}^R x d(\sqrt{R^2 - x^2}) - \sqrt{R^2 - x^2} dx \\ &= \int_{-R}^R \left(\frac{-x^2}{\sqrt{R^2 - x^2}} - \sqrt{R^2 - x^2} \right) dx = - \int_{-R}^R \frac{R^2 dx}{\sqrt{R^2 - x^2}} \end{aligned}$$

例 4. 设在原点放置正的点电荷 q , 求该点电荷所产生的电场在单位正电荷沿下述路径移动时所做的功:

(1) 沿空间直线 $\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-1}{0}$ 由点 $A(2, 0, 1)$ 到点 $B(1, 1, 1)$;

解: 原点处的正电荷对点 $M(x, y, z)$ 处的单位正电荷的作用力为 $\vec{F}(x, y, z) = kq \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$.

(1) 题设直线 L_1 的参数方程为

$$\begin{cases} x = 1 - t, \\ y = 1 + t, \quad t \in [-1, 0]. \\ z = 1, \end{cases}$$

于是所做的功为

$$\begin{aligned} W &= \int_{L_1(\overline{AB})} \vec{F} \cdot d\vec{\ell} = \int_{L_1(\overline{AB})} \frac{kqxdx}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{kqydy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \\ &\quad + \frac{kqzdz}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} = kq \int_{-1}^0 \frac{-(1-t) + (1+t)}{(3 + 2t^2)^{\frac{3}{2}}} dt \end{aligned}$$

第一、二类曲线积分之间的关系

设路径 $L \subset \mathbb{R}^3$ 是起点为 A , 终点为 B 的分段光滑曲线, 其参数方程为

$$\vec{\ell}(t) = (x(t), y(t), z(t)), \quad t \in [a, b],$$

而 $\vec{F} = (F_1, F_2, F_3) : L \rightarrow \mathbb{R}^3$ 为分段连续函数.

$\forall P \in L$, 设 L 在点 P 处的单位切向量为

$$\vec{\tau}^0(P) = (\cos \alpha(P), \cos \beta(P), \cos \gamma(P)).$$

于是 $\forall t \in [a, b]$, 我们有

$$\vec{\tau}^0(\vec{\ell}(t)) = \frac{\vec{\ell}'(t)}{\|\vec{\ell}'(t)\|} = \frac{(x'(t), y'(t), z'(t))}{\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}}.$$

由此立刻可得

$$\cos \alpha(\vec{\ell}(t)) = \frac{x'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}},$$

$$\cos \beta(\vec{\ell}(t)) = \frac{y'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}},$$

$$\cos \gamma(\vec{\ell}(t)) = \frac{z'(t)}{\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}}.$$

注意到 $d\ell = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}dt$, 故

$$x'(t)dt = \cos \alpha d\ell, \quad y'(t)dt = \cos \beta d\ell, \quad z'(t)dt = \cos \gamma d\ell.$$

进而我们就有

$$\int_{L(\widehat{AB})} \vec{F}(\vec{\ell}) \cdot d\vec{\ell} = \int_{L(\widehat{AB})} F_1(\vec{\ell})dx + F_2(\vec{\ell})dy + F_3(\vec{\ell})dz$$

$$= \int_a^b \left(F_1(\vec{\ell}(t))x'(t) + F_2(\vec{\ell}(t))y'(t) + F_3(\vec{\ell}(t))z'(t) \right) dt$$

$$= \int (F_1(x, y, z) \cos \alpha + F_2(x, y, z) \cos \beta + F_3(x, y, z) \cos \gamma) d\ell$$

注记

(A) 由于第二类曲线积分可以转化成第一类曲线积分, 因此只要不涉及到路径时, 第二类曲线积分就具有与第一类曲线积分类似的性质.

(B) 形式上, 我们有 $d\vec{\ell} = \vec{\tau}^0 d\ell$, 也即

$$dx = \cos \alpha d\ell, \quad dy = \cos \beta d\ell, \quad dz = \cos \gamma d\ell.$$

例 5. 求证: $\left| \int_L \vec{F}(\vec{\ell}) \cdot d\vec{\ell} \right| \leq \int_L \|\vec{F}(x, y, z)\| d\ell.$

证明: 由题设可知

$$\begin{aligned} \left| \int_L \vec{F}(\vec{\ell}) \cdot d\vec{\ell} \right| &= \left| \int_L (\vec{F} \cdot \vec{\tau}^0)(x, y, z) d\ell \right| \\ &\leq \int_L |(\vec{F} \cdot \vec{\tau}^0)(x, y, z)| d\ell \\ &\leq \int_L \|\vec{F}(x, y, z)\| d\ell, \end{aligned}$$

由此可知所证结论成立.

例 6. 求 $I = \oint_{L^+} \frac{-ydx + xdy}{x^2 + y^2}$, 其中 L 为圆周 $x^2 + y^2 = a^2$ ($a > 0$), 其正向为逆时针方向.

解: $\forall (x, y) \in L$, L 在该点的单位切向量为

$$\vec{\tau}^0(x, y) = \left(-\frac{y}{\sqrt{x^2 + y^2}}, \frac{x}{\sqrt{x^2 + y^2}} \right),$$

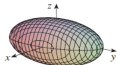
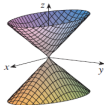
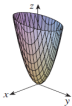
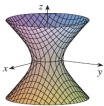
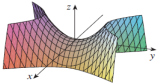
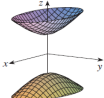
由此立刻可得

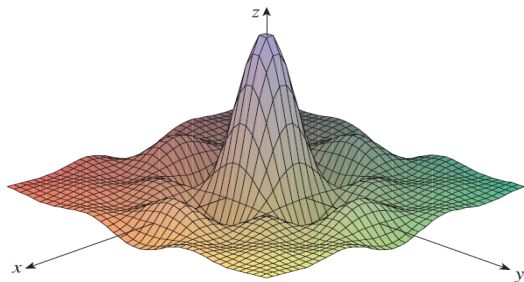
$$\begin{aligned}
 I &= \int_L \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right) \cdot \left(-\frac{y}{\sqrt{x^2 + y^2}}, \frac{x}{\sqrt{x^2 + y^2}} \right) \\
 &= \int_L \frac{d\ell}{\sqrt{x^2 + y^2}} = \int_L \frac{1}{a} d\ell = \frac{|L|}{a} = \frac{1}{a} \cdot 2\pi a = 2\pi.
 \end{aligned}$$

例 7. 计算 $I = \int_L (ye^x \sin y - xe^x \cos y) d\ell$, 其中 L 为圆周 $x^2 + y^2 = 1$ 位于第一象限的部分.

解: $\forall (x, y) \in L$, 圆周 L 在该点处的单位切向量为 $\vec{\tau}^0(x, y) = (-y, x)$, 于是我们有

$$\begin{aligned}
I &= \int_L (-e^x \sin y, -e^x \cos y) \cdot (-y, x) d\ell \\
&= \int_L (-e^x \sin y, -e^x \cos y) \cdot \vec{\tau}^0(x, y) d\ell \\
&= \int_{L^+} -e^x \sin y dx - e^x \cos y dy = - \int_{L^+} d(e^x \sin y) \\
&= -e^x \sin y \Big|_{(1, 0)}^{(0, 1)} = -\sin 1.
\end{aligned}$$

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ All traces are ellipses. If $a = b = c$, the ellipsoid is a sphere.	Cone 	$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x = k$ and $y = k$ are hyperbolas if $k \neq 0$ but are pairs of lines if $k = 0$.
Elliptic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	Hyperboloid of One Sheet 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c < 0$ is illustrated.	Hyperboloid of Two Sheets 	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ Horizontal traces in $z = k$ are ellipses if $k > c$ or $k < -c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.



(d) $f(x, y) = \frac{\sin x \sin y}{xy}$

同学们辛苦了!